

One factor random effects model

* Treatments are not fixed, but are randomly selected from a population of treatments.

Treatment effect $\tau_i \stackrel{iid}{\sim} N(0, \sigma_\mu^2)$

$$Y_{ij} = \underbrace{\mu}_{\text{fixed}} + \underbrace{\tau_i}_{\text{random}} + \underbrace{\epsilon_{ij}}_{\text{random noise}} \quad \epsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2)$$

$\{\tau_i\}$ and $\{\epsilon_{ij}\}$ are indep.

$$Y_{ij} \sim N(\mu, \sigma_\mu^2 + \sigma^2)$$

$$\text{Cov}(Y_{ij}, Y_{ij'}) = \sigma_\mu^2 \text{ for } j \neq j'$$

$$\text{Cor}(Y_{ij}, Y_{ij'}) = \frac{\sigma_\mu^2}{\sigma^2 + \sigma_\mu^2} = \rho \leftarrow \text{intraclass correlation coefficient}$$

$$1 \geq \rho \geq 0$$

* Inference for σ^2

* Inference for $\sigma_\mu^2 \rightarrow H_0: \sigma_\mu^2 = 0$ vs.

* Inference for $\rho \quad H_A: \sigma_\mu^2 > 0$

If the design is balanced, i.e., $n_i = n$ for all $i = 1, \dots, r$

then $\frac{\text{MSTR}}{\sigma^2 + n\sigma_\mu^2} \sim \frac{\chi_{r-1}^2}{r-1}$

indep
(check)

$$\frac{\text{MSE}}{\sigma^2} \sim \frac{\chi_{n_T - r}^2}{n_T - r} \quad (n_T = nr)$$

[Write in terms of ϵ_{ij} 's]

$$\Rightarrow F = \frac{\text{MSTR}}{\text{MSE}} \sim \left(\frac{\sigma^2 + n\sigma_\mu^2}{\sigma^2} \right) F_{r-1, n_T - r}$$

$$\text{MSTR} = \frac{\text{SSTR}}{r-1}$$

$$\text{SSTR} = n \sum_{i=1}^r (\bar{Y}_{i.} - \bar{Y}_{..})^2$$

$$\begin{pmatrix} \bar{Y}_{1.} \\ \vdots \\ \bar{Y}_{r.} \end{pmatrix} \sim N\left(\underline{\mu}, \left(\sigma_\mu^2 + \frac{\sigma^2}{n}\right) I\right)$$

Reject $H_0: \sigma_\mu^2 = 0$ vs. $H_A: \sigma_\mu^2 > 0$ at level α if

$$F^{obs} > F_{r-1, n_T-r}^{1-\alpha}$$

Confidence interval for ρ :

$$\rho = \frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma^2} = \frac{\sigma_\mu^2 / \sigma^2}{1 + \sigma_\mu^2 / \sigma^2}$$

$$F = \frac{MSTR}{MSE} \sim \left(1 + n \frac{\sigma_\mu^2}{\sigma^2}\right) F_{r-1, n_T-r}$$

Let $\theta = \frac{\sigma_\mu^2}{\sigma^2}$ then $\rho = \frac{\theta}{1+\theta}$

$$\frac{F}{1+n\theta} \sim F_{r-1, n-r}$$

$$P\left(F_{r-1, n_T-r}^{\alpha/2} \leq \frac{F}{1+n\theta} \leq F_{r-1, n_T-r}^{1-\alpha/2}\right) = \alpha$$

$\Rightarrow$ $(1-\alpha)$ CI for θ is

$$\frac{1}{n} \left(\underbrace{\frac{F}{F_{r-1, n_T-r}^{\alpha/2}}}_{L} - 1 \right) \leq \theta \leq \frac{1}{n} \left(\frac{F}{F_{r-1, n_T-r}^{1-\alpha/2}} - 1 \right) \underbrace{\quad}_U$$

$\Rightarrow$ $(1-\alpha)$ CI for $\rho = \frac{\theta}{1+\theta}$ is $\left[\frac{L}{1+L}, \frac{U}{1+U} \right]$

Another
~~Alternative~~ expression for $(1-\alpha)$ CI for ρ

$$\tilde{L} = \frac{MSE}{MSTR} F_{r-1, n_T-r}^{\alpha/2}$$

$$\tilde{U} = \frac{MSE}{MSTR} F_{r-1, n_T-r}^{1-\alpha/2}$$

The CI for ρ $[L^*, U^*]$

$$L^* = \frac{1 - \tilde{U}}{1 + (n-1)\tilde{U}}$$

$$U^* = \frac{1 - \tilde{L}}{1 + (n-1)\tilde{L}}$$

Note: $\rho = 0$ iff $\sigma_\mu^2 = 0$

So if $0 \in$ confidence interval for ρ , then this signifies that we cannot reject $H_0: \sigma_\mu^2 = 0$

In the example of interviews conducted by personal offices

$$MSE = 73.3$$

$$MSTR = 394.9$$

$$\left. \begin{matrix} n = 4 \\ r = 5 \end{matrix} \right\} n_T = 20$$

$$\text{Recall } F = \frac{MSTR}{MSE} = 5.39$$

$$F_{4,15}^{0.95} = 3.06$$

So we rejected $H_0: \sigma_\mu^2 = 0$
at 5% level

$$\alpha = 0.1 \Rightarrow 90\% \text{ CI for } \rho \text{ turns to be } [0.160, 0.885]$$

Confidence interval for σ_μ^2 (only under balanced design i.e. $n_i = n$)

Unbiased estimate of σ_μ^2 is $s_\mu^2 = \frac{1}{n}(MSTR - MSE)$

$$\left[\begin{array}{l} E(MSTR) = \sigma^2 + n\sigma_\mu^2 \\ E(MSE) = \sigma^2 \end{array} \right]$$

Note: s_μ^2 can be negative
So it is a common practice to set $\hat{\sigma}_\mu^2 = (s_\mu^2)_+$ ← positive part

Satterthwaite procedure (for finding approximate confidence interval for σ_μ^2)

$$\begin{aligned} \theta &= \sum_{i=1}^k c_i E M_i \quad \text{where } M_i \stackrel{\text{indep}}{\sim} \theta_i \cdot \frac{\chi_{d_i}^2}{d_i} \\ &= \sum_{i=1}^k c_i \theta_i \end{aligned}$$

Then $\hat{\theta} = \sum_{i=1}^k c_i M_i$ Finding approximate distribution of $\hat{\theta}$

unbiased est.

$\hat{\theta} \approx \theta \frac{\chi_D^2}{D}$ where $D = \frac{\left(\sum_{i=1}^k c_i M_i\right)^2}{\sum_{i=1}^k \frac{(c_i M_i)^2}{d_i}}$

(D can be fractional)

$\chi_k^2 \equiv \text{Gamma}\left(\frac{k}{2}, \frac{1}{2}\right)$

Using this procedure, an approximate $(1-\alpha)$ CI for σ_μ^2 becomes

$$\left[\frac{D \times s_\mu^2}{\chi_D^2(1-\alpha/2)}, \frac{D \times s_\mu^2}{\chi_D^2(\alpha/2)} \right]$$

where $D = \frac{(n s_\mu^2)^2}{\frac{(MSTR)^2}{r-1} + \frac{(MSE)^2}{n_T - r}}$

For the example (personal offices)

$s_\mu^2 = \frac{1}{4} \left(\underset{\text{MSTR}}{394.9} - \underset{\text{MSE}}{73.3} \right) = \underline{80.4}$ $n=4$
 $r=5$

$D = 2.63$

$\chi_{2.63}^2(0.05) = 0.245$

$\chi_{2.63}^2(0.95) = 7.163$

90% approx CI for $\sigma_\mu^2 = [29.5, 863.1]$

Prediction problem:

If τ_i 's were fixed effects, subject to identifiability condition $\sum n_i \tau_i = 0$, then the BLUE for τ_i is

$$\hat{\tau}_i = \bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot}$$

Under the random effects model, $\tau_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_\mu^2)$

{ Information on $\{\tau_i\}$ from the data is captured through the conditional distribution of τ_i given $\underline{Y}$ }

Furthermore, due to independence of $\{(\tau_i, \{\overset{\text{data}}{\varepsilon_{ij}}\}_{j=1}^{n_i})\}_{i=1}^r$ the conditional distⁿ of τ_i given $\underline{Y}$ is the same as the conditional distⁿ of τ_i given $\underline{Y}_i = \begin{pmatrix} Y_{i1} \\ \vdots \\ Y_{in_i} \end{pmatrix}$

and $(\tau_i, \underline{Y}_i)$ is jointly normal

$$\Rightarrow \tau_i | \underline{Y}_i \sim \mathcal{N}(\quad)$$

Define the Best Linear Unbiased Predictor of τ_i to be the conditional mean of τ_i given $\underline{Y}$

$$\begin{aligned} g_i(\underline{Y}) &= E(\tau_i | \underline{Y}) = E(\tau_i | \underline{Y}_i) \\ &= \left(\frac{n_i \sigma_y^2}{n_i \sigma_y^2 + \sigma_\mu^2} \right) (\bar{Y}_{i\cdot} - \mu) \end{aligned} \quad \left| \begin{array}{l} \text{"Best" refers to the} \\ \text{fact that} \\ g_i(\underline{Y}) = \underset{g}{\operatorname{argmin}} E(\tau_i - g(\underline{Y}))^2 \end{array} \right.$$

$\hookrightarrow$ shrinkage factor

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix} \right), \quad E(X|Y) = \mu_x + \Sigma_{xy} \Sigma_{yy}^{-1} (Y - \mu_y)$$

$\mu_y = \mu \mathbf{1}_n$ $\Sigma_{yy} = \sigma^2 I + \sigma_\mu^2 \mathbf{1} \mathbf{1}^T$
 $\mu_x = 0$ $\Sigma_{xy} = \sigma_\mu^2 \mathbf{1} \mathbf{1}^T$

Under balanced design ($n_i = n$)

$\sigma^2 + n\sigma_\mu^2$ is estimated by MSTR

σ^2 is estimated by $\hat{\sigma}^2 = \text{MSE}$

μ is estimated by $\bar{Y}_{..}$

So, the EBLUP of τ_i is $\frac{nS_\mu^2}{nS_\mu^2 + \hat{\sigma}^2} (\bar{Y}_{i.} - \bar{Y}_{..})$

↑
estimated best linear
unbiased predictor

where $S_\mu^2 = \frac{1}{n} (\text{MSTR} - \text{MSE})$